

Sakil Sarker

Applied Machine Learning Researcher

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Portfolio | GitHub | LinkedIn | LeetCode

Professional Summary

Applied ML Researcher / AI Engineer (MSc, AI) specializing in **LLMs, Retrieval-Augmented Generation (RAG), transformers**, and **explainable** systems that turn research into reliable products. Shipped **transformer/NLP** features to production (cloud-native, monitored, low-latency) and presented/published at **ACM/IEEE, MobileHCI, and CUI**. Strengths: **quality–latency trade-offs**, evaluation at scale, **adaptive UX for learning**, and clean deployment on **AWS + Docker**.

Skills

Languages: Python, C/C++, JavaScript, SQL, ABAP

ML/AI: PyTorch, TensorFlow, Transformers, Hugging Face, LLMs, RAG, NLP, Summarization, Prompt Engineering, XAI/Explainability

MLOps: Docker, Git, CI/CD, Experiment Tracking (Weights & Biases / MLflow), Monitoring, Drift Detection, A/B Testing

Data/Cloud: AWS (S3, Lambda, ECS/EKS, SageMaker), Firebase, SLURM, Linux, REST APIs

Practices: Reproducible Research, Documentation, Code Reviews

Experience

Graduate Research Assistant — Ontario Tech University, Oshawa, ON Sep 2024 – Present

- Engineered **RAG** pipelines with hierarchical summarization and **layered explainability** to produce auditable, trustworthy answers in learning tools.
- Built an **adaptive reading assistant** that detects confusion/engagement signals (highlights, scroll, dwell) and triggers targeted AI help; improved section completion in pilot runs.
- Ran **quality vs. latency** benchmarks across transformer backbones; standardized evaluation artifacts that guided model and deployment decisions.

Graduate Teaching Assistant — Ontario Tech University, Oshawa, ON Sep 2024 – Present

- Led AI/ML, Big Data, and Data Mining labs; authored **debug playbooks** and starter kits that reduced support load and sped up assignment iterations.

Software Engineer (AI/ML & SAP ABAP) — BJIT Group, Dhaka, Bangladesh Oct 2021 – Aug 2024

- Productized **NLP/predictive** features in SaaS (React/Node on AWS) with monitoring and safe rollback; hardened releases and reduced incidents.
- Automated **SAP Fiori/ABAP** workflows to remove manual steps and variance in back-office operations across enterprise deployments.
- Recognized as a **top performer** in company-wide SAP ABAP/Fiori training for hands-on proficiency and on-time delivery.

Deep Learning Research Intern — Carnegie Mellon University, Pittsburgh, PA May 2021 – Oct 2021

- Trained DL models for **3D protein structure** prediction; analyzed error modes and iterated architectures that informed subsequent research.

- Consolidated experiments into a **reproducible** codebase adopted by collaborators.

Education

MSc in Computer Science (AI Specialization) — Ontario Tech University Sep 2024 – Present
GPA: 4.3/4.3

- Research in applied LLMs, generative AI, and human-AI collaboration.
- Thesis: AI-driven platform for collaborative academic reading.

BSc in Computer Science and Engineering — East West University Apr 2017 – Jun 2021
Grade: 87%

- Dean's Scholarship recipient (2019–2021); graduated top 5% of class.

Publications

- **PaperPal: AI-Powered Research Comprehension Assistant** *Accepted, ACM CUI 2025.*
Focus: LLM-driven assistance for research reading and comprehension in collaborative settings.
Preprint: arXiv (to appear)
- **Focusing on the Unfocused: Small-Scale Non-Profit Interaction Study** *ACM COMPASS 2023.*
DOI: 10.1145/3605150
- **K-Cosine-Means Clustering Algorithm** *IEEE ICECIT 2021.*
DOI: 9641480
- **K-Cosine-Medoids Clustering Algorithm** *IEEE EICT 2021.*
DOI: 9733540

Talks & Presentations

- **Presenter, ACM MobileHCI** Egypt | 2025.
SwipeSense: Exploring the Feasibility of Back-of-Device Swipe Interaction Using Built-In IMU Sensors. Results + demo; sensing pipeline, signal features, and mobile interaction implications.
- **Poster Presenter, ACM CUI** University of Waterloo, Canada | 2025.
PaperPal: AI-Powered Research Comprehension Assistant. LLM-driven reading support; interaction flow, layered explainability, and early user feedback.

Representative Projects

- **LitSense: AI-Powered Collaborative Reading Platform** — successor to *PaperPal* (ACM CUI 2025).
 Unified LLM summarization, **layered explainability**, and collaborative annotation; demoed **adaptive help** triggered by reading signals.
YouTube (Demo): <https://youtu.be/3oBVuGn2N-o>
- **Real-Time Spam Detection (Transformer-based).**
 Designed for **low p95 latency** and stable precision under shift; added an evaluation harness for drift monitoring.
GitHub: Advanced-Email-Spam-Ham-Detection-System-Using-Transformer-Based-Model
- **AI-Powered Recruitment Advisor.**
 LLM-driven resume parsing and question generation for fair, inclusive screening.
GitHub: AI-Powered-Recruitment-Advisor

Certifications & Awards

- AWS Certified Cloud Practitioner 2023
- SAP Certified Associate – ABAP (NetWeaver 7.50) 2022
- TCPS-2 CORE Ethics Certification (Canada) 2024
- Dean's Scholarship, East West University 2019–2021
- Top 2 in SAP ABAP/Fiori Training, BJIT 2021